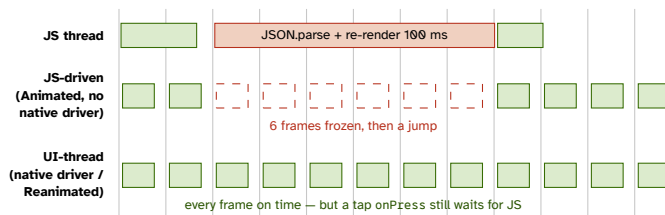
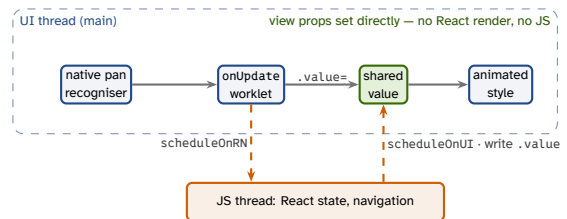


**In one line:** Smooth animation means computing each frame **on the UI thread**, so a busy JS thread cannot drop it. Animated's `useNativeDriver` does this for a pre-declared animation of *transform/opacity*; **Reanimated** runs your own *worklets* on the UI thread; **Gesture Handler** recognises touches natively and feeds them. iOS analogue: Core Animation keeps running while your code blocks.

JS thread busy for 100 ms — what the user sees (one box = one 16.7 ms frame)



Gesture → worklet → style, never touching JS



## Animated (core)

- `Animated.Value` (held in `useRef`) · `ValueXY`; drivers timing (duration + easing), spring (physical, interruptible), decay (velocity); compose sequence/parallel/stagger/loop.
- `interpolate({inputRange, outputRange, extrapolate: 'clamp'})` — one value drives many props (rotation, colour, opacity).
- **`useNativeDriver: true`** serialises the animation graph to native *once*; frames then run on UI. Only **non-layout** props: `transform`, `opacity`. **Not** width, height, top, margin, flex — they need Yoga layout every frame.
- `Animated.event(..., {useNativeDriver:true})` binds scroll offset natively — the classic collapsing header.
- **LayoutAnimation:**  
`configureNext(Presets.easeInEaseOut)` before a state change animates the *next layout* — cheap, global, no gestures, not interruptible. Old-arch Android:  
`UIManager.setLayoutAnimationEnabledExperimental(true)`.

## Reanimated — how it wins

- A **worklet** ('worklet' directive, added by the Babel plugin for hook callbacks) is copied into a **second JS runtime on the UI thread**; captured variables are copied, not shared.
- `useSharedValue` — readable/writable from both runtimes via `.value`; writing it does **not** re-render React.
- `useAnimatedStyle(fn)` re-runs on UI when a shared value it reads changes and patches the native view — **any** prop, layout included.
- `withTiming`, `withSpring`, `withDecay`, `withSequence`, `withRepeat`, `withDelay`; `useDerivedValue`; `useAnimatedScrollHandler`; `cancelAnimation`.
- `scheduleOnRN(fn, ...args)` UI → JS; `scheduleOnUI(fn, ...args)` JS → UI (react-native-worklets; 3x `runOnJS(fn)(args)` / `runOnUI` are deprecated re-exports in 4). Async hops — at the end of a gesture, never per frame.
- **Layout animations:** `entering={FadeIn}`, `exiting={FadeOut}`, `layout={LinearTransition}` on `Animated.View`.
- Reanimated 4 requires the New Architecture; worklets moved to `react-native-worklets`.

## Example — swipe card, springs back

```
const x = useSharedValue(0);
const pan = usePanGesture({
  onUpdate: (e) => { x.value = e.translationX; }, // RNGH 3
  onDeactivate: (e) => {
    if (Math.abs(e.translationX) > 120) scheduleOnRN(dismiss);
    else x.value = withSpring(0, { velocity: e.velocityX });
  },
});
const style = useAnimatedStyle(() => ({
  transform: [{ translateX: x.value }],
  opacity: interpolate(Math.abs(x.value), [0, 200], [1, 0.3],
    Extrapolation.CLAMP),
}));
return <GestureDetector gesture={pan}>
  <Animated.View style={style} /></GestureDetector>;
```

## Gesture Handler 3 (hooks; 2.x still maintained)

- **Native recognisers** (`UIGestureRecognizer` on iOS), replacing the JS responder system — they work while JS is busy. Root: `GestureHandlerRootView`.
- A config object per hook: `usePanGesture({...})`, `useTapGesture`, `useLongPressGesture`, `usePinchGesture`, `useRotationGesture`, `useFlingGesture` → `<GestureDetector gesture={g}>`. Callbacks `onBegin` · `onActivate` · `onUpdate` · `onDeactivate` (v2 `Gesture.Pan().onStart` / `onEnd`; `onChange` folded into `onUpdate`).
- `useSimultaneousGestures(pinch, rotate)` — both active · `useExclusiveGestures(doubleTap, tap)` — first that succeeds, in order · `useCompetingGestures` (v2 Race). Across components: `requireToFail` (UIKit `require(toFail:)`), `simultaneousWith`, `block`.
- Hook-API callbacks run on the **UI thread** by default (Reanimated integration); `runOnJS: true` moves them to JS.

## Traps

- ! "Native driver makes everything smooth" — not layout props, and a JS `onPress/setState` is still late.
- ! Calling a plain JS function from a worklet without `scheduleOnRN` crashes; reading `sv.value` in render gives a stale value. `scheduleOnRN(setState, v)` per frame rebuilds the JS bottleneck.
- ! Animating width/height costs a layout pass per frame even on UI — prefer `transform: scale` where the look allows.

## Remember

**Frames on UI, decisions on JS:** gesture → worklet → shared value → style; cross to JS only at the end.